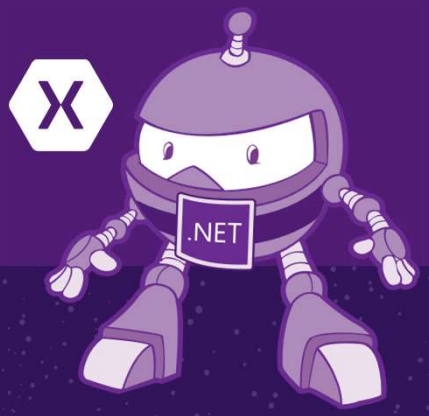

.NET Conf
"Focus on Xamarin"
.NET
focus.dotnetconf.net

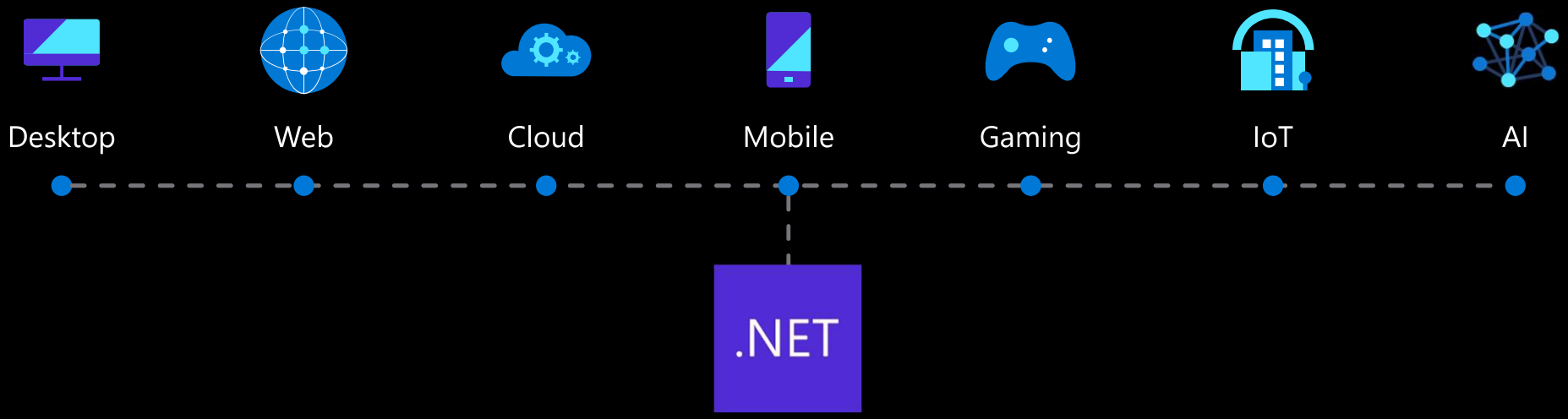
Xamarin All the Things

Amanda Silver, CVP Program Management, DevDiv
David Ortinau, Principal Program Manager
Maddy Leger, Program Manager

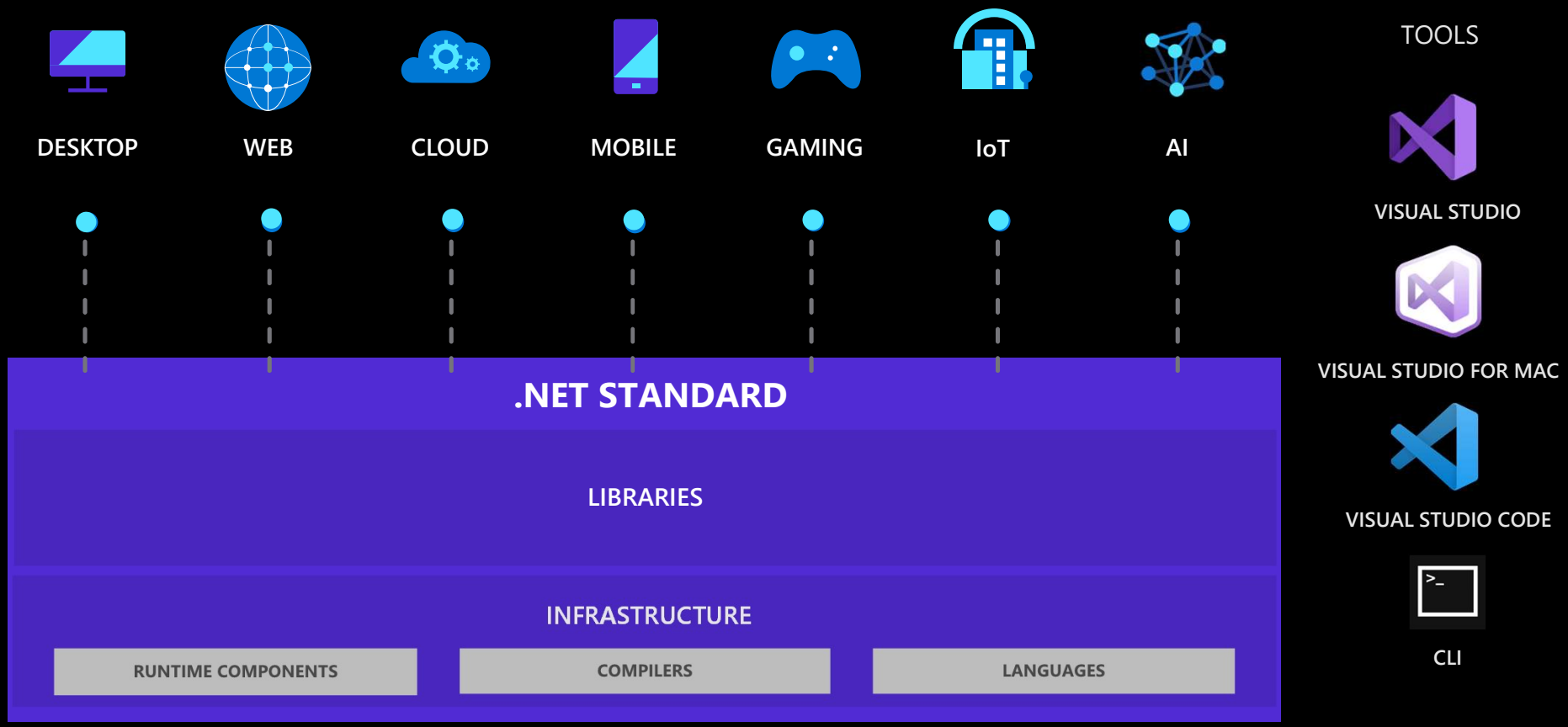


.NET

Your platform for building anything



.NET – A unified platform



.NET ecosystem momentum

+1M
New .NET devs in last year

in Visual Studio

#1 Most Loved
Framework

.NET Core

stackoverflow

Top 30
Highest velocity OSS projects

github.com/dotnet
github.com/aspnet

.NET foundation

Top 5
Language on GitHub

C#



7x
Faster than Node.js

ASP.NET Core

TechEmpower

40%
New to .NET are students

dot.net download survey

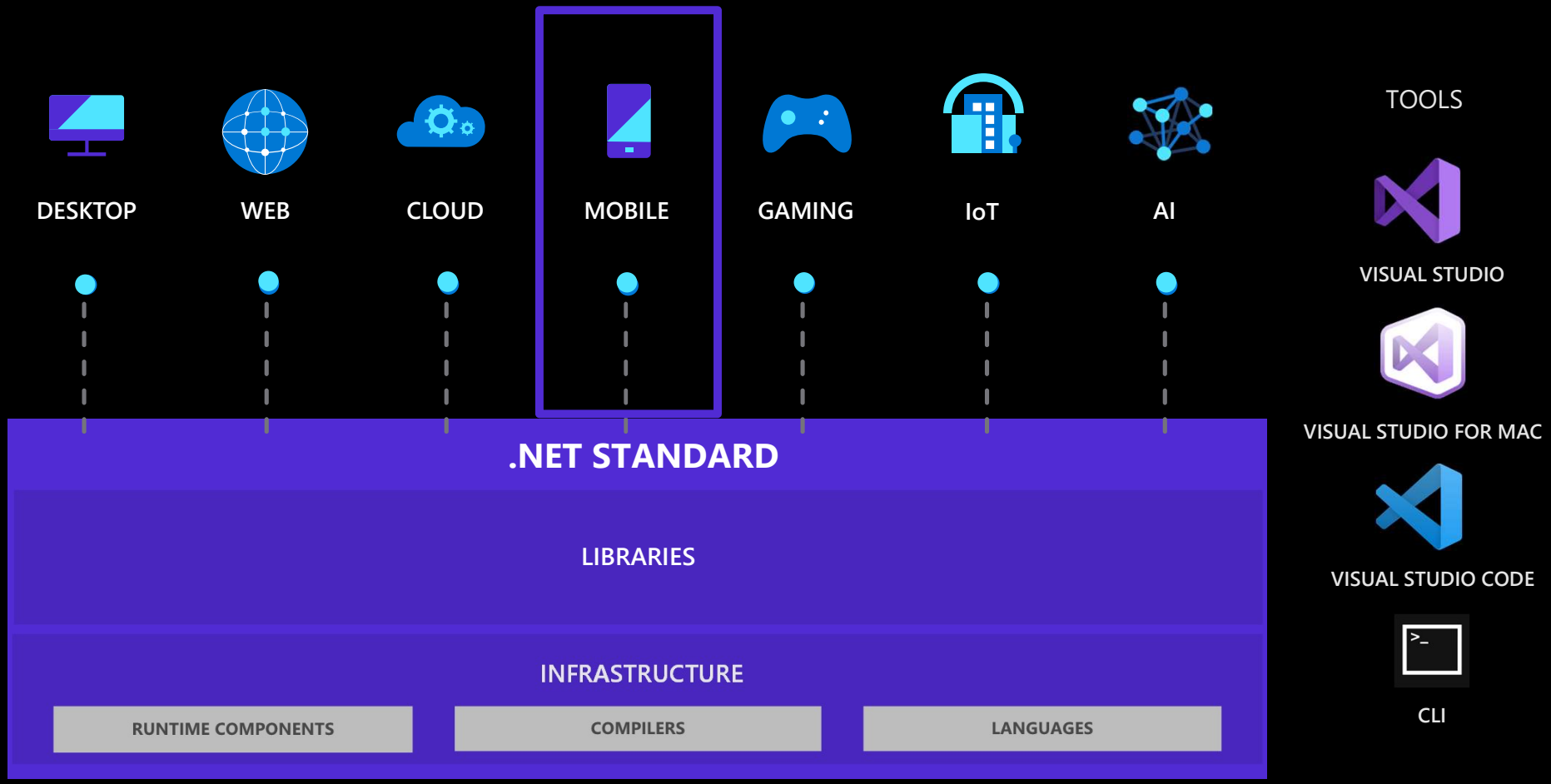
.NET open source momentum



87%
contributors outside Microsoft

110,000+
accepted PRs from community

.NET – A unified platform



With Xamarin, you can build fast, beautiful, native apps in less time, with less code, within a thriving ecosystem

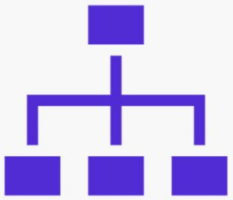
.NET mobile apps with Xamarin



iOS & Android with C#

Native performance, native APIs, native UI

Build native mobile apps with Visual Studio



Cross-Platform

Build cross-platform UI with Xamarin.Forms

Access native APIs from shared code with Xamarin.Essentials

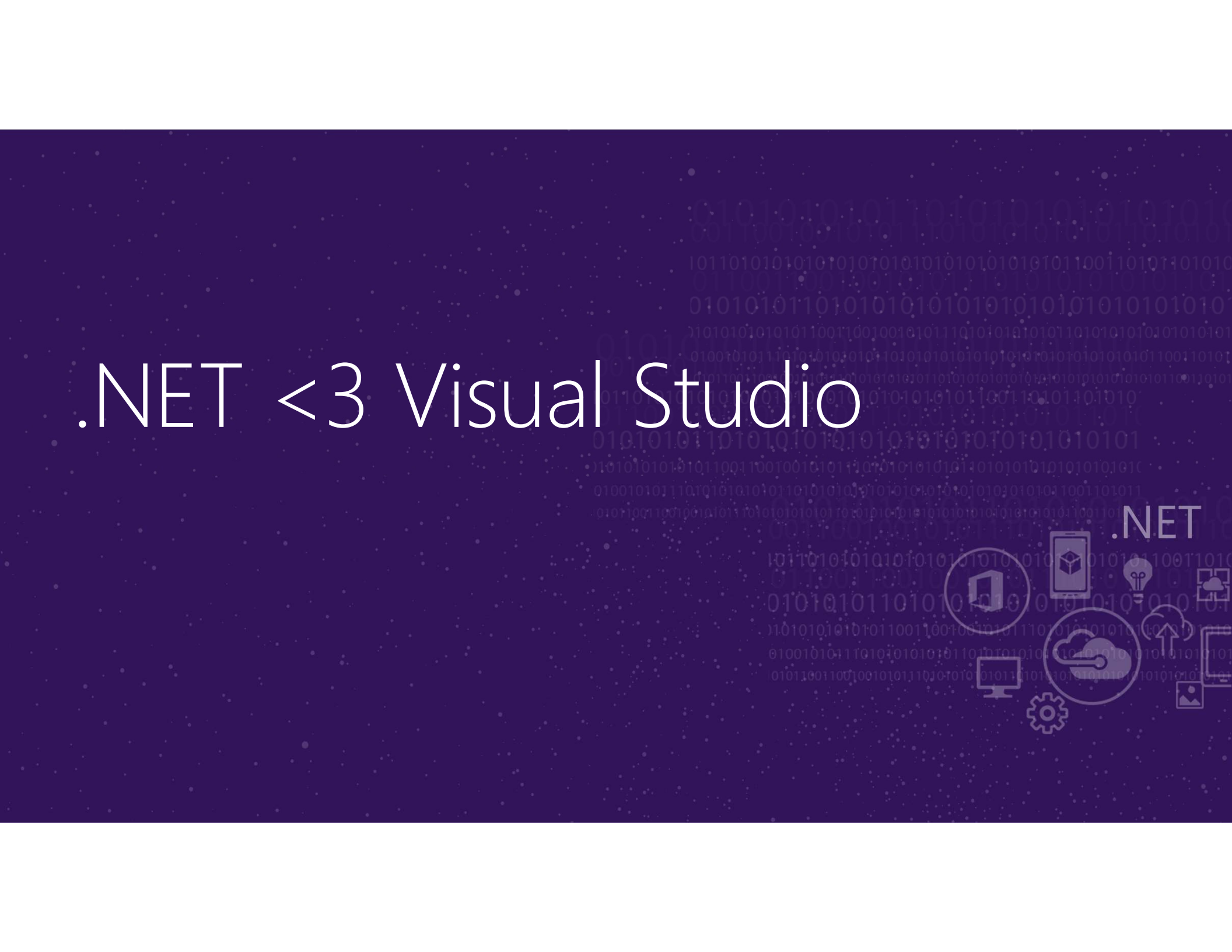
Share code with .NET Standard



Free & Open Source

Xamarin is part of .NET and open source on GitHub

.NET <3 Visual Studio



Visual Studio 2019 key focus areas

Personal and Team Productivity

Focus on what's important

Faster startup & install time

New search experience

Improve your productivity

Improved debugging

CodeLens for all

Refactoring and code cleanup

Collaborate across teams

Visual Studio Live Share

Azure Repos pull requests in IDE

Modern Development Support

Support modern app types

.NET mobile with Xamarin, including Hot Reload and Hot Restart

.NET Core development

Cloud native, serverless, containers

Be cloud-ready with Azure

Local/offline development

Secure by default (Key Vault)

Develop and publish without leaving the IDE

Constant Innovation

Get started faster and easier

New start window

Git-first workflow

Improved new project dialog

Seamlessly upgrade

Smart assistance while you code

Visual Studio IntelliCode, including Android XML and Xamarin.Forms XAML

Production-environment debugging

Xamarin is now over 2x faster to install



Visual Studio 2017



Visual Studio 2019

RELEASED

Xamarin Developer Productivity

XAML Hot Reload

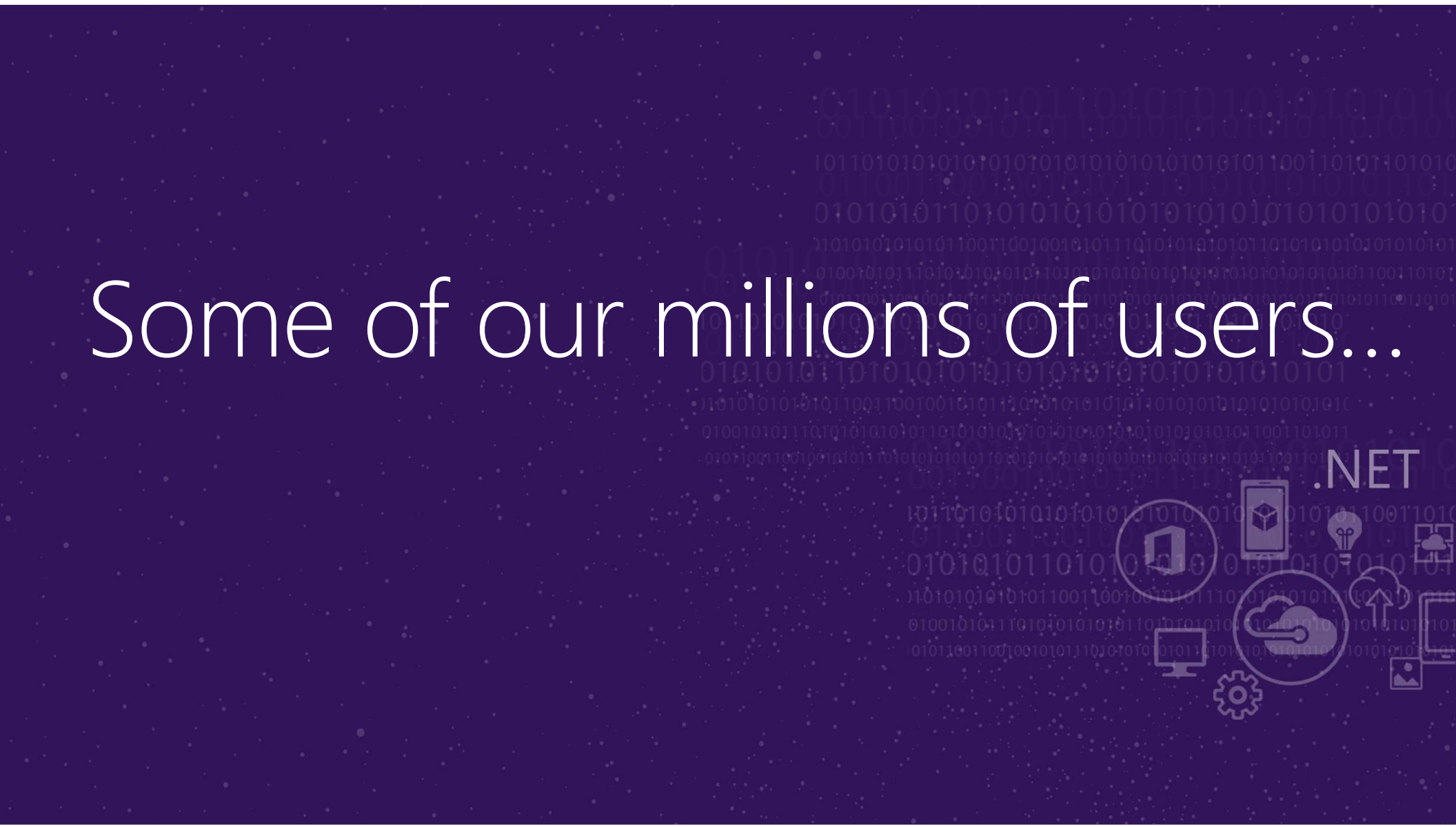
Iterate quickly on your Xamarin.Forms UIs while debugging

Hot Restart

Quickly redeploy your Xamarin.Forms app without a full compile

Visual Studio version 16.5

Some of our millions of users…





Xamarin

Loved by developers,
trusted by enterprise



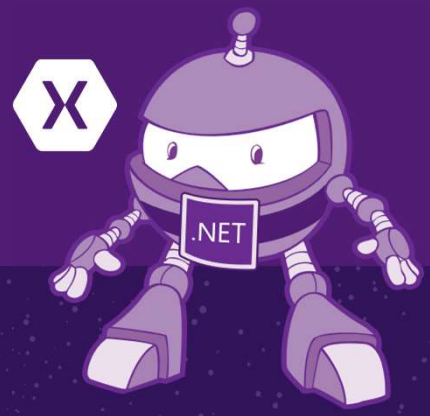


UPS paves the way for better service with faster mobile development using Xamarin

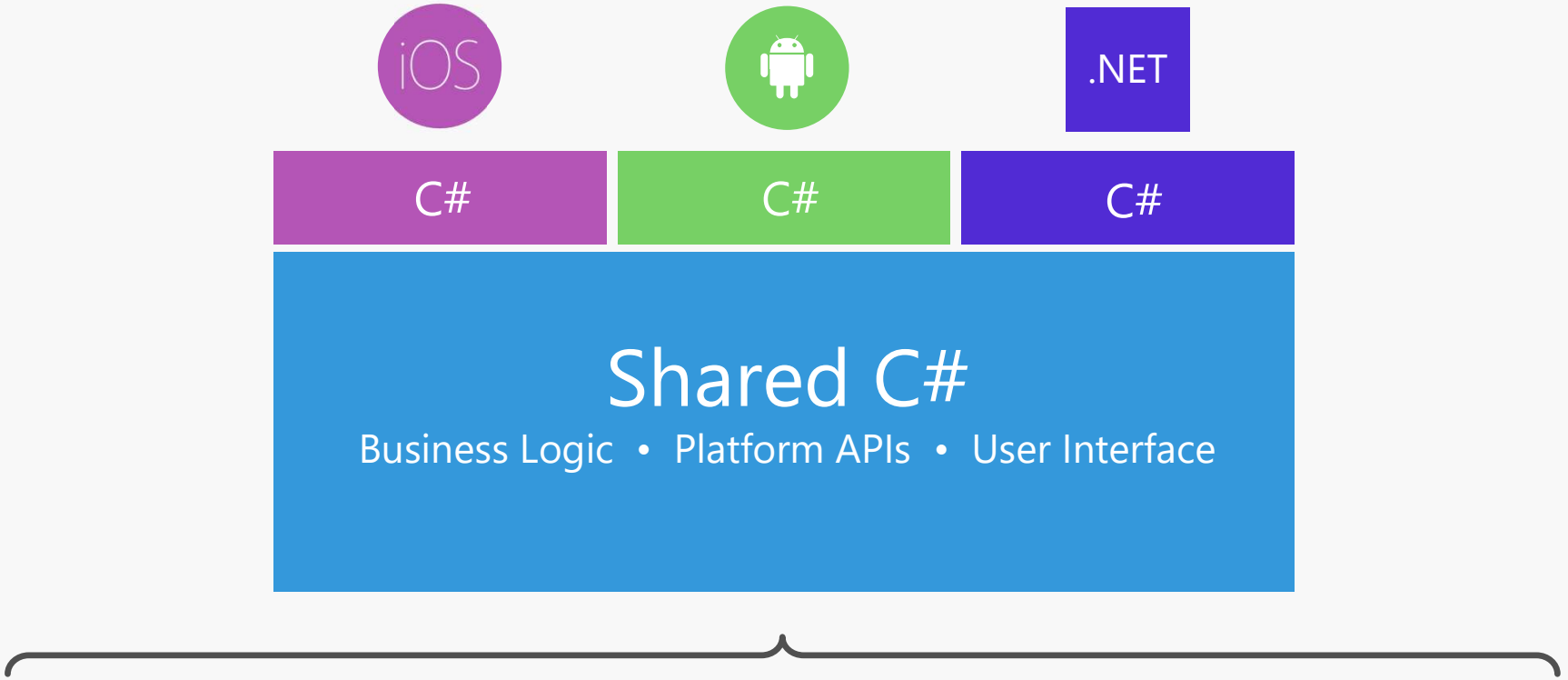
David Ortinau
Maddy Leger



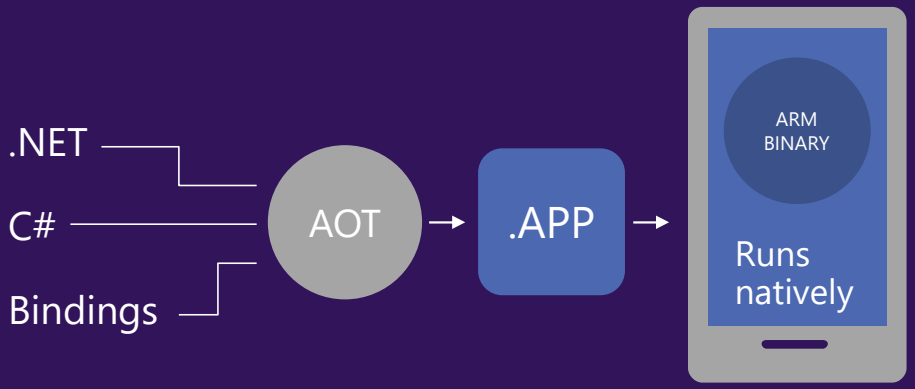
Fast, beautiful, and native



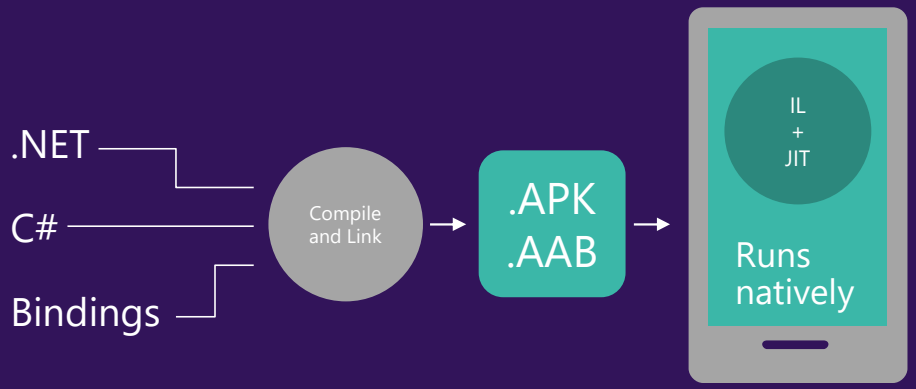
Xamarin App Architecture



Native execution, native performance

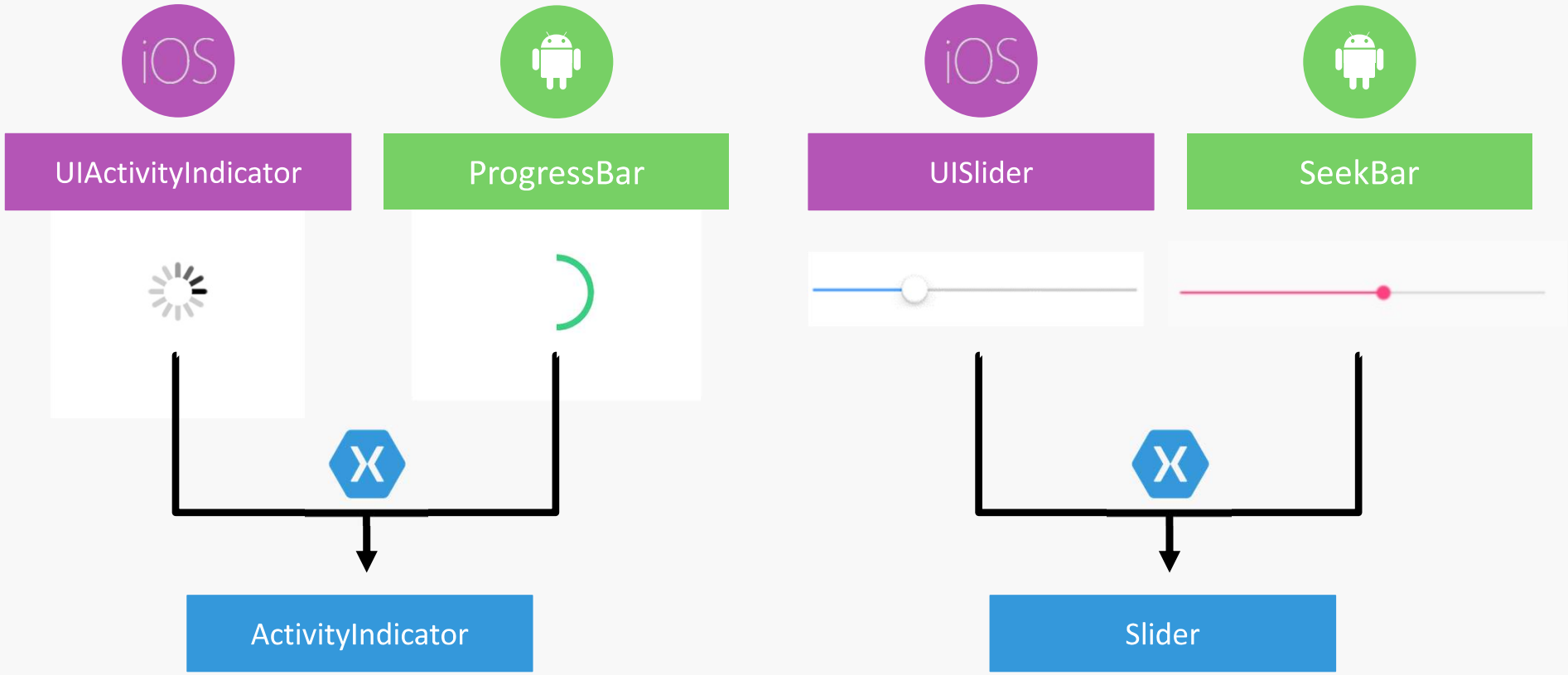


Xamarin.iOS does full Ahead Of Time (AOT) compilation to produce an ARM binary for Apple's App Store.



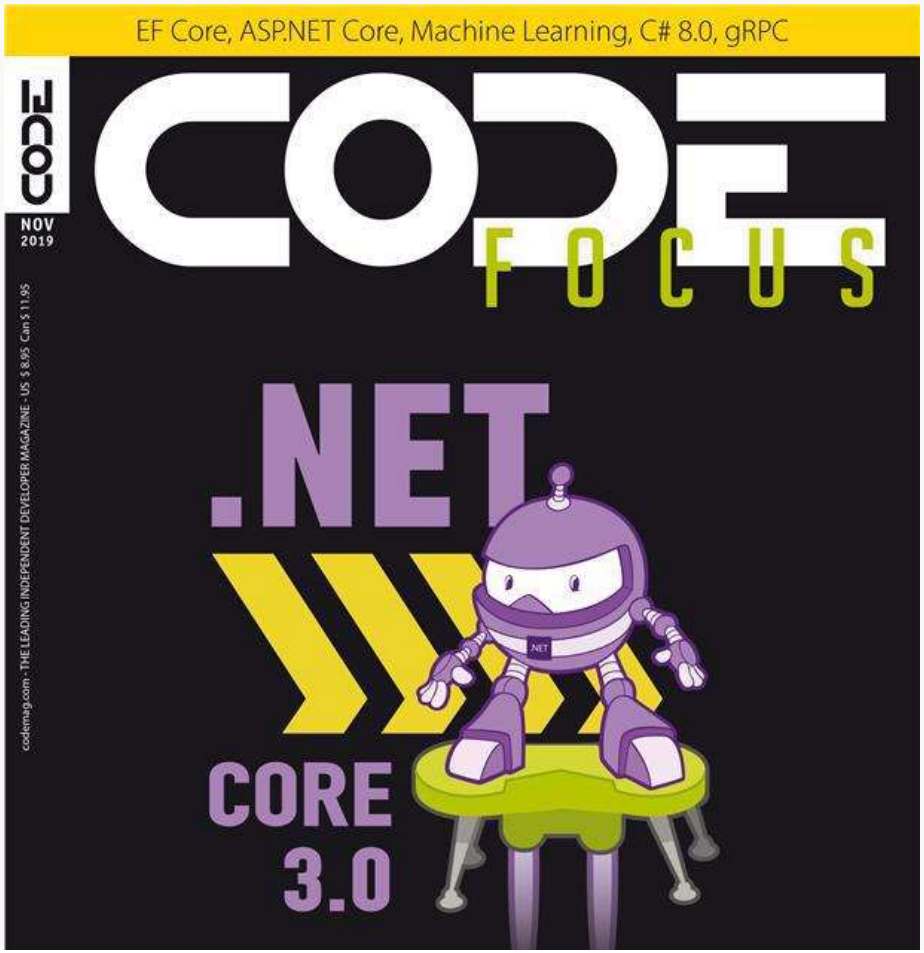
Xamarin.Android takes advantage of Just In Time (JIT) compilation on the Android device.

Native UI, even when shared

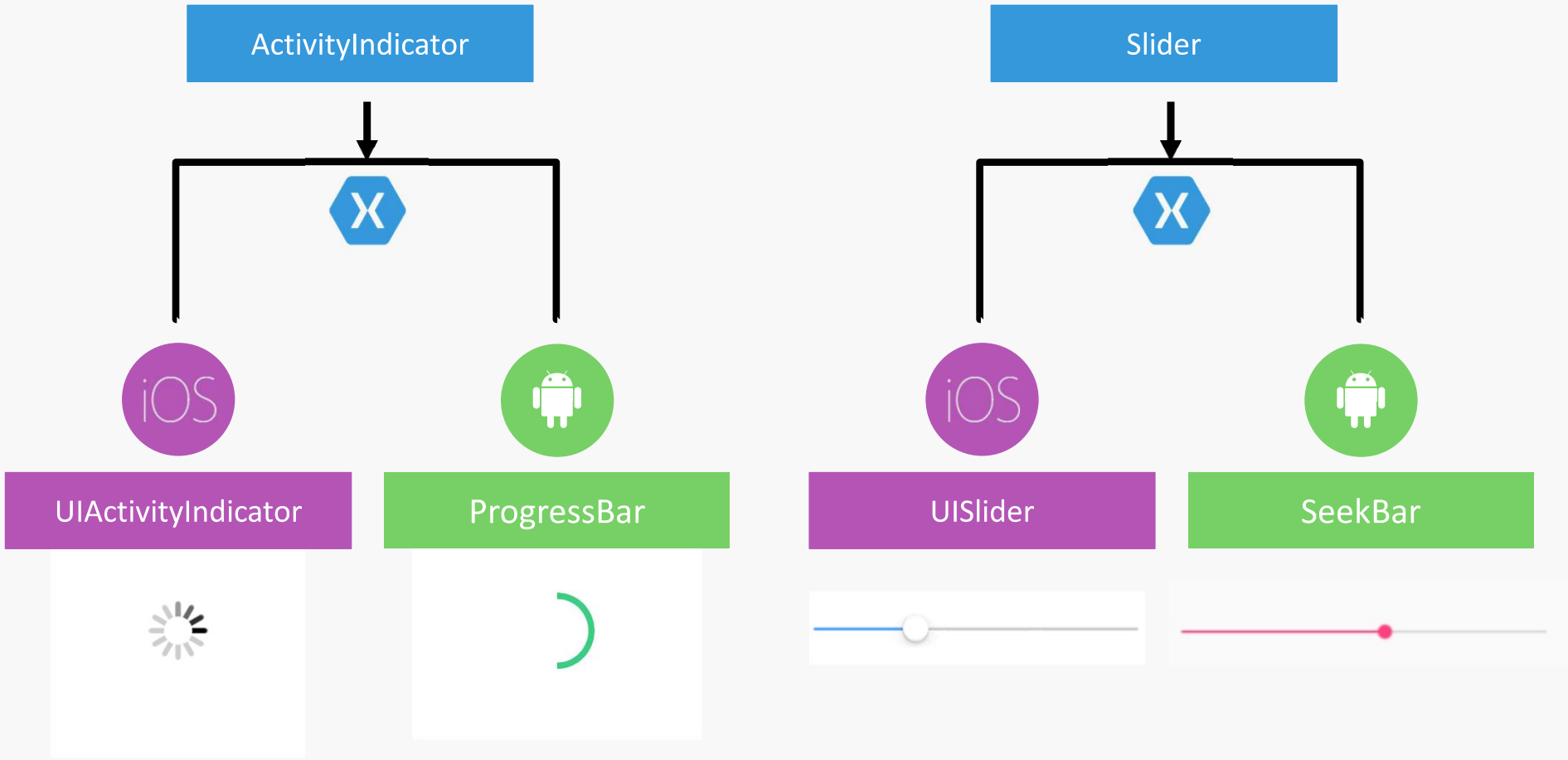


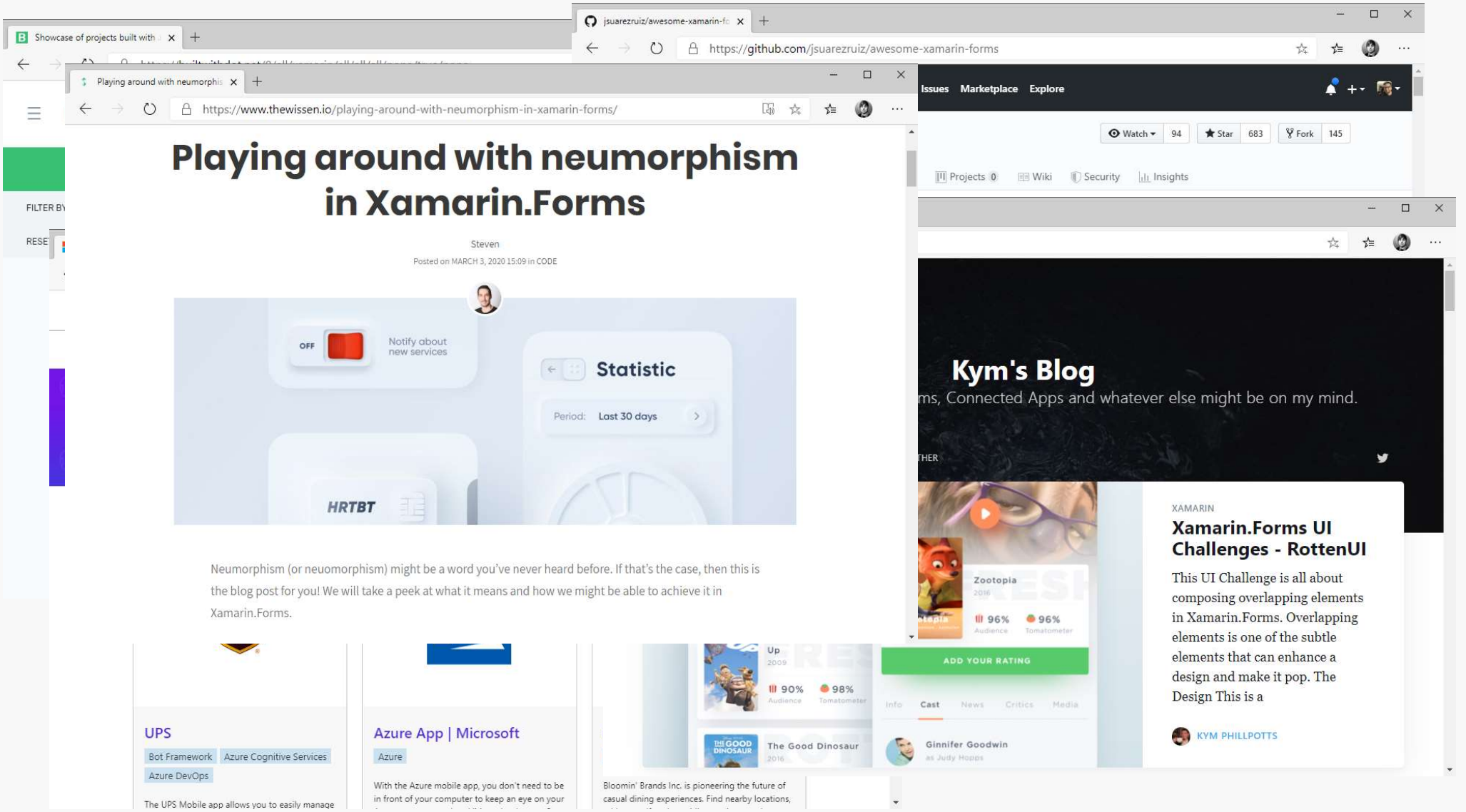
Native Forms: Shared UI in Native Apps

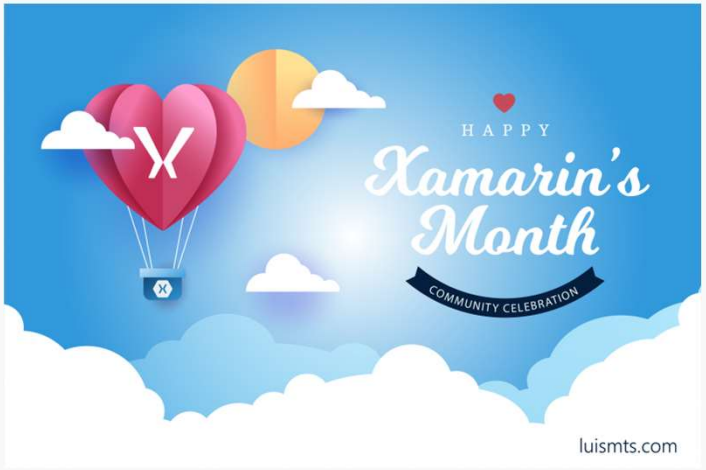




Native UI, even when shared

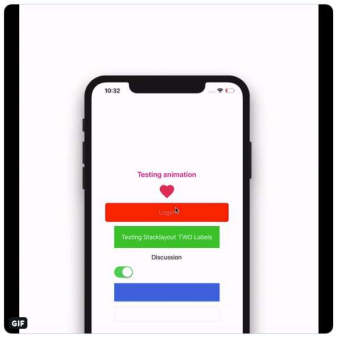



Showcase of projects built with
jsuarezruiz/awesome-xamarin-fo
https://github.com/jsuarezruiz/awesome-xamarin-forms
Playing around with neumorphis
https://www.thewissen.io/playing-around-with-neumorphism-in-xamarin-forms/
Issues Marketplace Explore
Watch 94 Star 683 Fork 145
Projects 0 Wiki Security Insights
Playing around with neumorphism in Xamarin.Forms
Steven
Posted on MARCH 3, 2020 15:09 in CODE
OFF
Notify about new services
Statistic
Period: Last 30 days
HRTBT
Neumorphism (or neuomorphism) might be a word you've never heard before. If that's the case, then this is the blog post for you! We will take a peek at what it means and how we might be able to achieve it in Xamarin.Forms.
Kym's Blog
ms, Connected Apps and whatever else might be on my mind.
THER
XAMARIN
Xamarin.Forms UI Challenges - RottenUI
This UI Challenge is all about composing overlapping elements in Xamarin.Forms. Overlapping elements is one of the subtle elements that can enhance a design and make it pop. The Design This is a
KYM PHILLPOTTS
Zootopia
96%
96%
ADD YOUR RATING
Info Cast News Critics Media
Ginnifer Goodwin
as Judy Hopps
Up
90%
98%
The Good Dinosaur
UPS
Bot Framework
Azure Cognitive Services
Azure DevOps
The UPS Mobile app allows you to easily manage
Azure App | Microsoft
Azure
With the Azure mobile app, you don't need to be in front of your computer to keep an eye on your
Bloomin' Brands Inc. is pioneering the future of casual dining experiences. Find nearby locations,



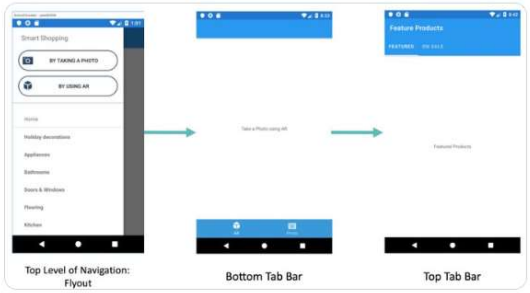
Charlin Agramonte
@Chard003

I just wrote a new article about Plug and Play animations in Xamarin Forms
xamgirl.com/plug-and-play-... #xamarinmonth #Xamarin @xamarinhq



Alejandro Ruiz ➡ SOON @XamarinSummit 2019 BR
@alejandroruizva

Xamarin Month: Fall in Love with Xamarin.Forms Shell #Xamarin #xamarinmonth
alejandroruizvarela.blogspot.com/2019/02/xamari...



John Taubensee
@jtaubensee

Just wrote a short post on how to add an ActivityIndicator to a #XamarinForms Button.
taubensee.net/activitybutton... #XamarinMonth



Daniel Monettelli L.
@DanielMonetelli

Xamarin.Forms Update - Los enamorados CSS y BindableLayout - #DanielMonetelli #xamarinmonth #XamarinForms #CSS #BindableLayout #AdobeXD #UI #UX



Martin Zikmund #UnoConf
@mzikmunddev

10 lovely things you should tell your friends about #Xamarin #xamarinmonth #day4 #XamarinForms #appdev #unoplatform #mvvmcross
blog.mzikmund.com/2019/02/things...



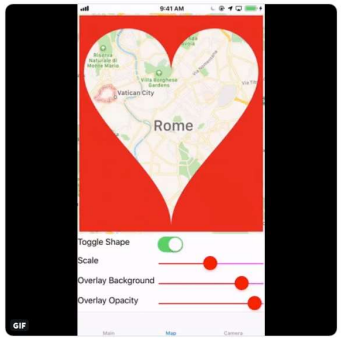
Marco Siccardi
@msicc

Use NuGets for your common #Xamarin (Forms) code (and automate the creation process) #XamarinMonth
bit.ly/2GciX04



Rendy Del Rosario
@rdelrosario

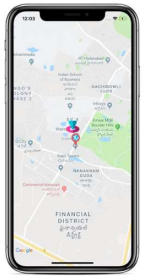
I just published a post about creating an Overlay view in Xamarin Forms xamboy.com/2019/02/23/ove... @xamarinhq #xamarinforms #xamarinmonth



Aritra Das
@aritra_das

I just wrote a new article, titled "Using custom markers and Latitude Longitude bounds in Google Maps on Xamarin Forms"
gameofthreads.online/custom-markers...

#XamarinForms #xamarinmonth #GoogleMaps #Android #iOS @xamarinhq @PlanetXamarin @davidortinau



pujolsluis
@pujolsluis1

Hey Guys! just posted my article for the #xamarinmonth initiative about my recent experience helping out a member of our Xamarin Forms Dev Group on Facebook, she loved Xamarin and Prism @Elsa_AM4Y that now we got her first Xamarin Post for the community🥰🙌



Hussain Abbasi
@HussainNAbbasi

New Blog Post: Stunning Animations In With Lottie.
My contribution to #XamarinMonth @ @xamarinhq #XamarinForms #Lottie



Luis Beltran
@darkicebeam

Just published "#Azure Table Storage with #Xamarin Forms" on my blog as part of the #xamarinmonth initiative:

luisbeltran.mx/2019/02/02/azu...

#MVPBuzz



Luis Matos
@luismatosluna

The future of #XamarinForms, an opinion of what may come in the following updates. Join me to see the Visual improvements in the development UI that #xamarin will have for us.

Post: bit.ly/2IzGOIV #XamarinMonth #XamarinUniverse #VisualControls @xamarinhq



Matthew Soucoup
@CodeMillMatt

I wrote a special blog as part of #xamarinmonth!

5 Azure Offerings I ❤️For Xamarin Development!

codemilltech.com/5-azure-offeri...

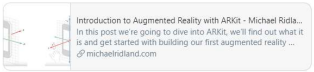
Kirk Patrick Junsay
@KPjunsay

Profiling memory leaks in Xamarin Forms :)
codesandchips.blog/2019/02/22/pro... #xamarinmonth

Michael Ridland
@mrid00z

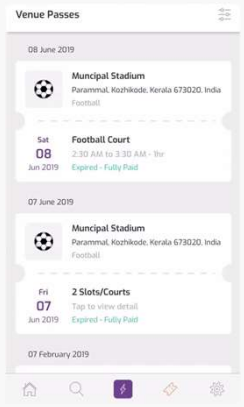
Replying to @luismatosluna

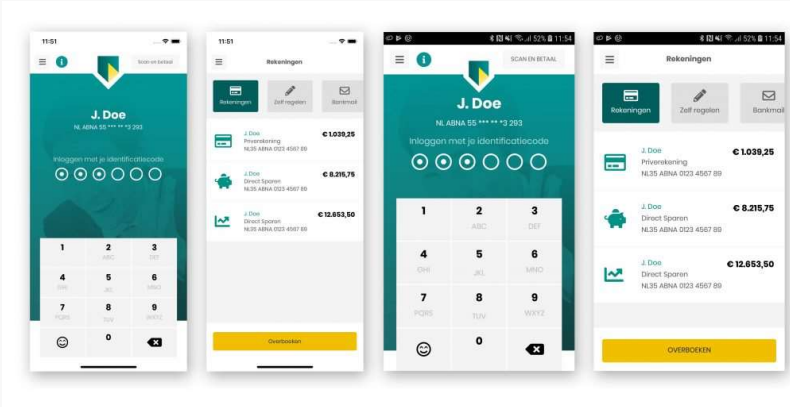
Yep, michaelridland.com/mobile/introdu... #xamarinmonth Introduction to Augmented Reality with ARKit



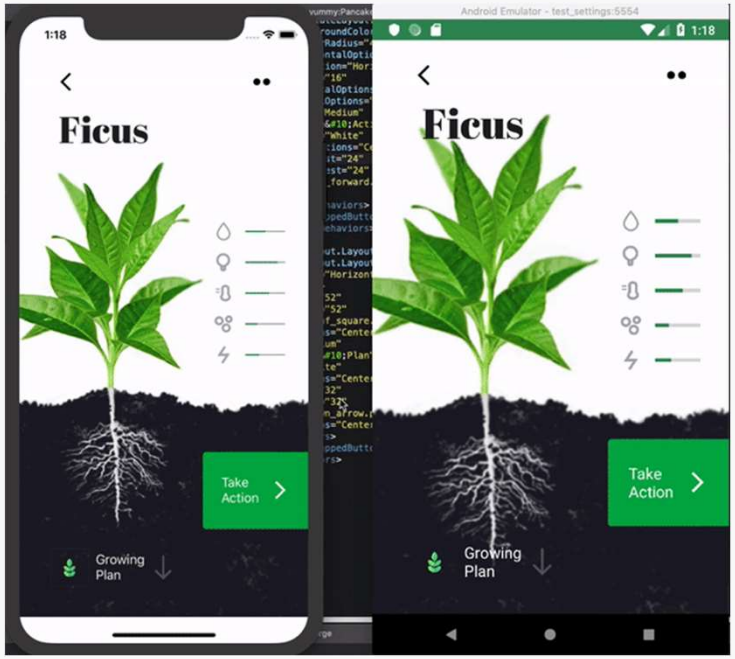
Sebastian | tsjdev-apps.de
@tsjdevapps

I just published Use FontAwesome in a app link.medium.com/1wyqcZyTjU #xamarin #xamarinmonth

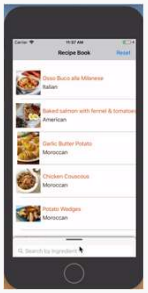

Venue Passes
08 June 2019
Muncipal Stadium
Parammal, Kozhikode, Kerala 673020, India
Football
Sat
08
Jun 2019
Football Court
2:30 AM to 3:30 AM - 1hr
Expired - Fully Paid
07 June 2019
Muncipal Stadium
Parammal, Kozhikode, Kerala 673020, India
Football
Fri
07
Jun 2019
2 Slots/Courts
Tap to view detail
Expired - Fully Paid
07 February 2019

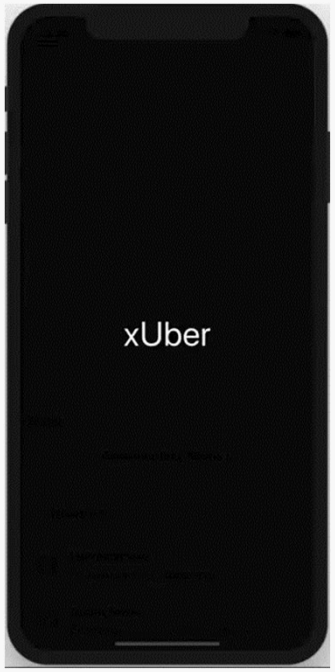

J. Doe
Inloggen met je identificatiecode
Rekeningen
OVERBOEKEN

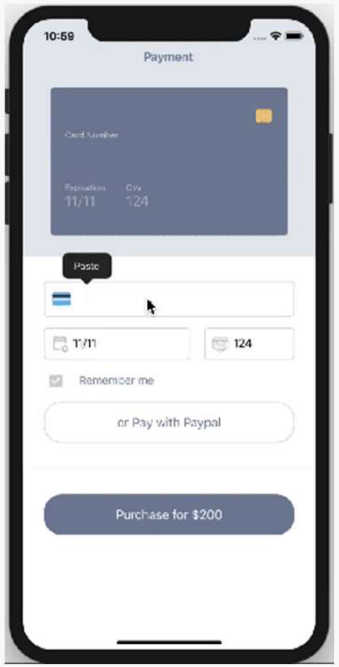

FEATURED BLOG
#XAMARINUIJULY


Ficus
Take Action
Growing Plan


Sandro Botticelli
March 1, 1445 - May 17, 1510
Highlight Paintin
The birth of Venus
Primavera


Recipe Book


xUber


Payment
Paste
Remember me
or Pay with Paypal
Purchase for $200


Run Away! Sunday, July 14, 2019
Google
Time (hrs)
1:14
Calories
853
Distance
8.37
Average
159
Average
6.7

Awesome Xamarin Component Vendors









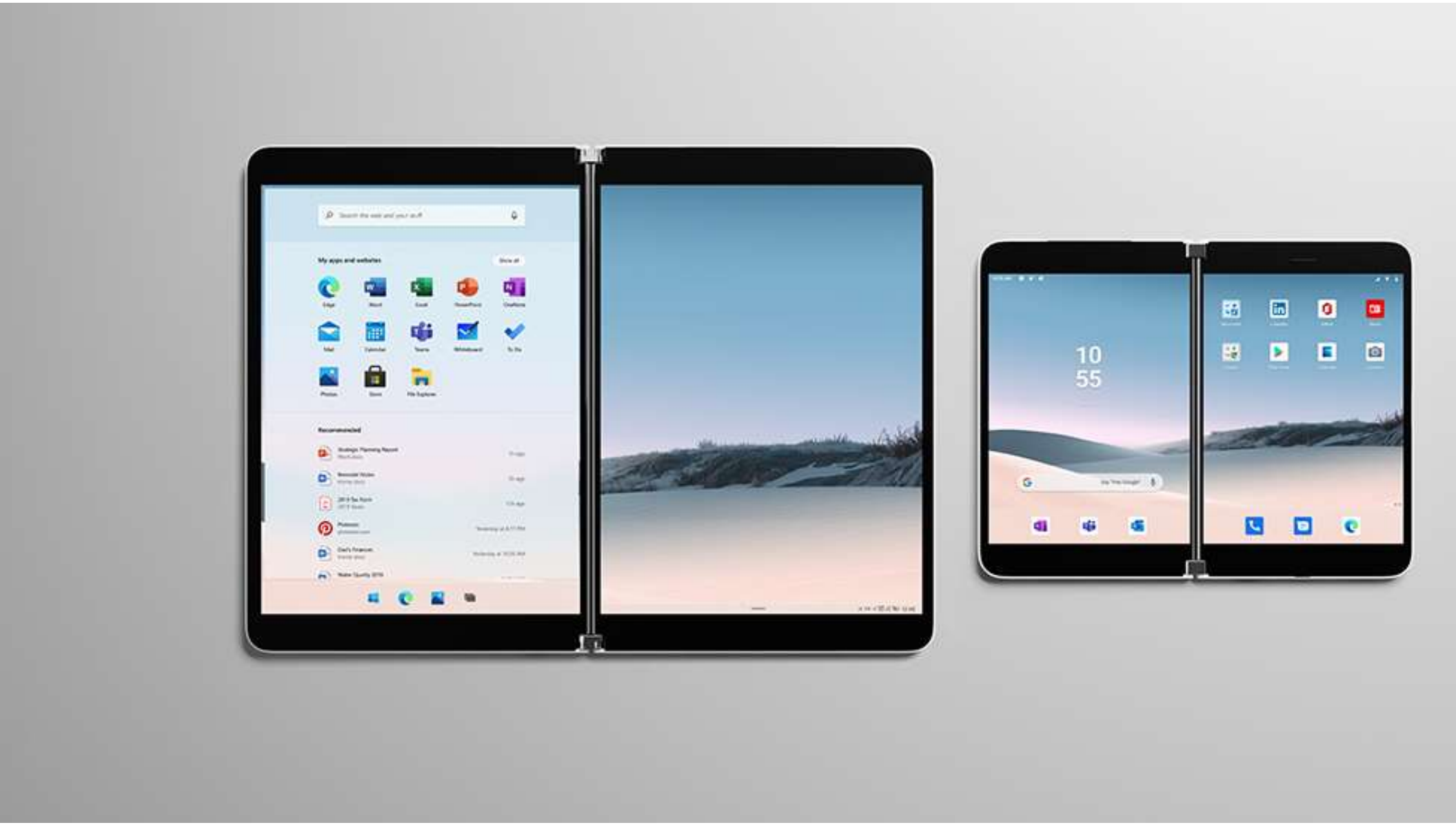


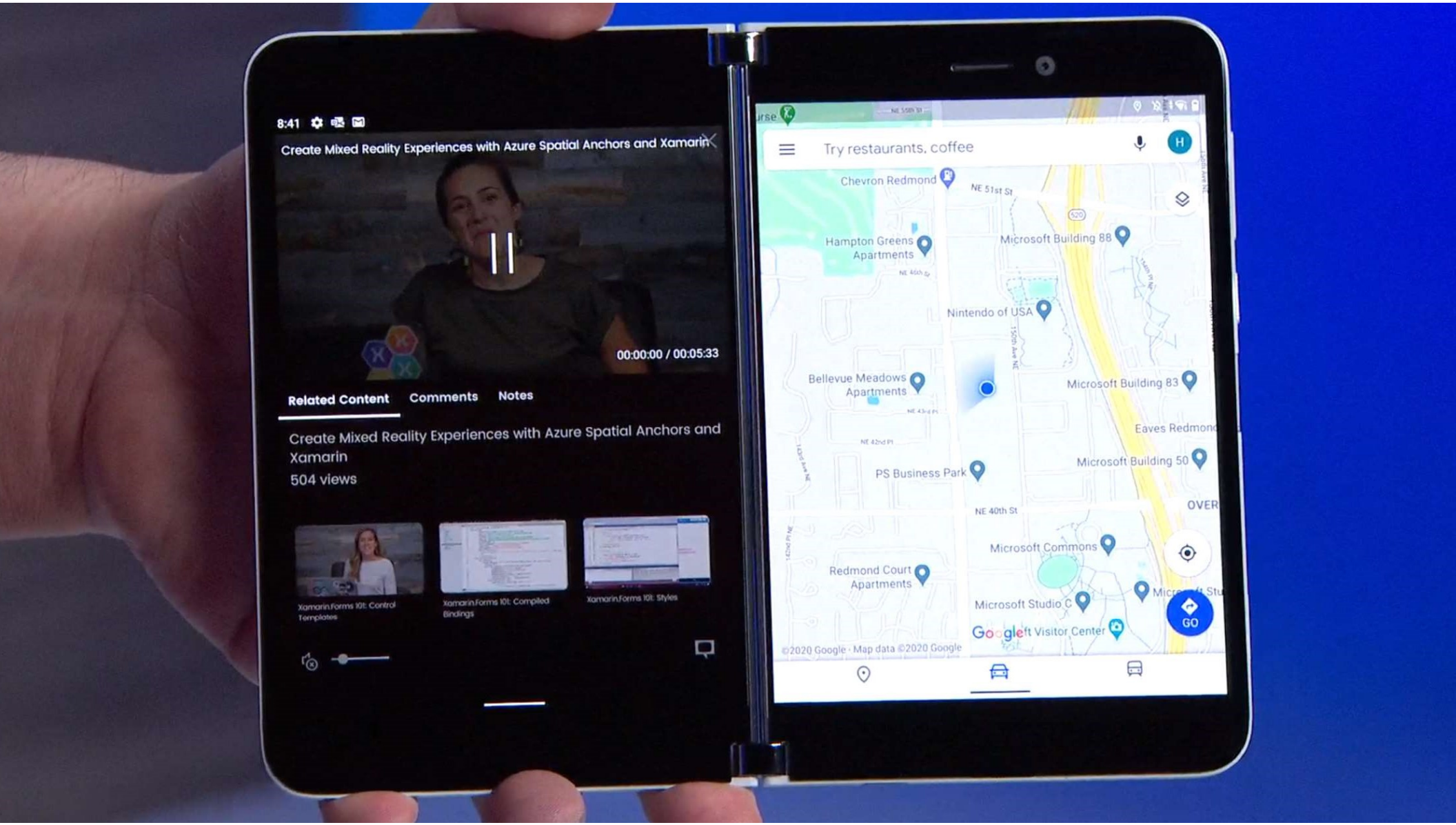




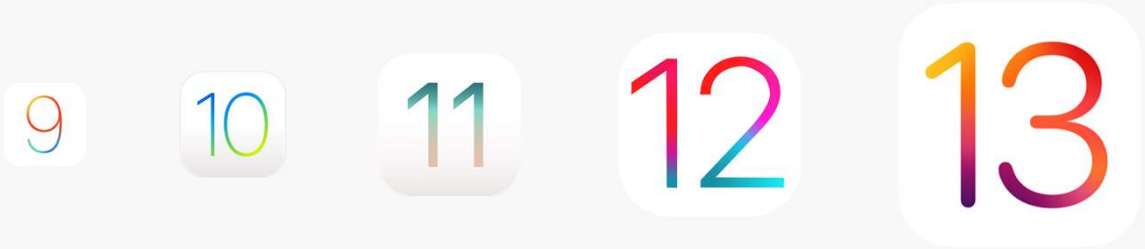





10
55


8:41
Create Mixed Reality Experiences with Azure Spatial Anchors and Xamarin
00:00:00 / 00:05:33
Related Content
Comments
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Create Mixed Reality Experiences with Azure Spatial Anchors and Xamarin
504 views
Xamarin.Forms 101: Control Templates
Xamarin.Forms 101: Compiled Bindings
Xamarin.Forms 101: Styles
Try restaurants, coffee
Chevron Redmond
NE 51st St
Hampton Greens Apartments
Microsoft Building 88
Nintendo of USA
Bellevue Meadows Apartments
Microsoft Building 83
Microsoft Building 50
PS Business Park
NE 40th St
Microsoft Commons
Redmond Court Apartments
Microsoft Studio C
Visitor Center
GO
©2020 Google · Map data ©2020 Google

In Step and Up to Date



Xcode 11.4 support
out now



Android 11 support
coming to Xamarin
in 16.6 Preview 2

Android App Bundles

- Specific Languages
- Proper Resource Screen Densities
- Best Match ABIs





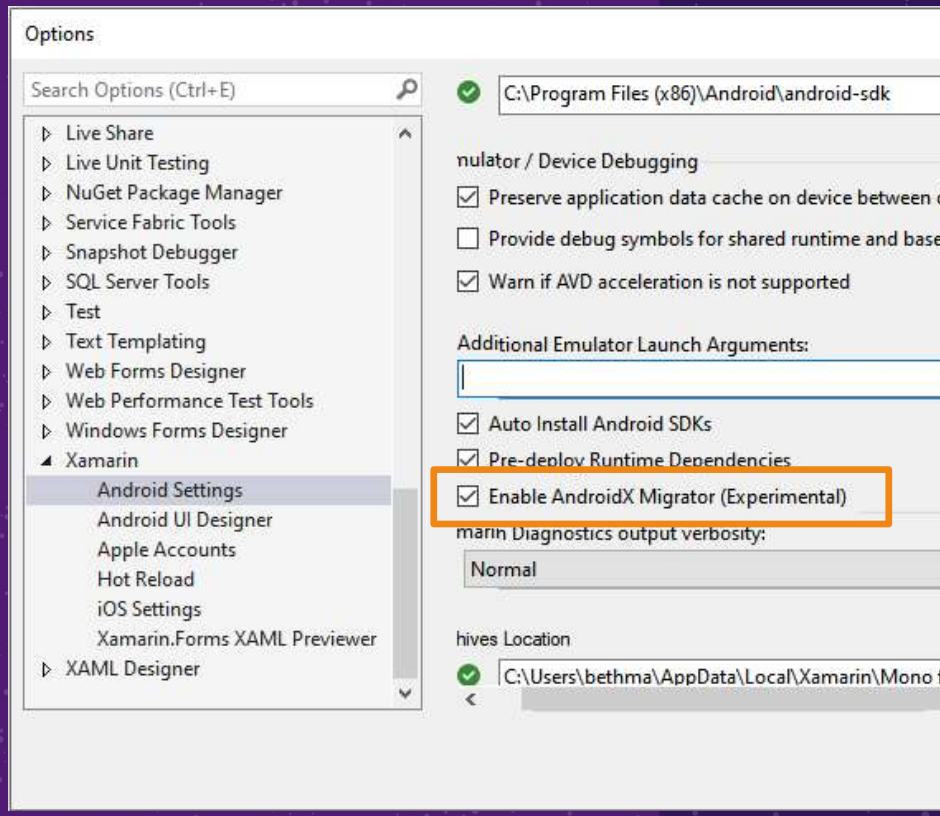
AndroidX



PREVIEW

AndroidX Migrator

Migrate your Android projects seamlessly



Visual Studio version 16.5

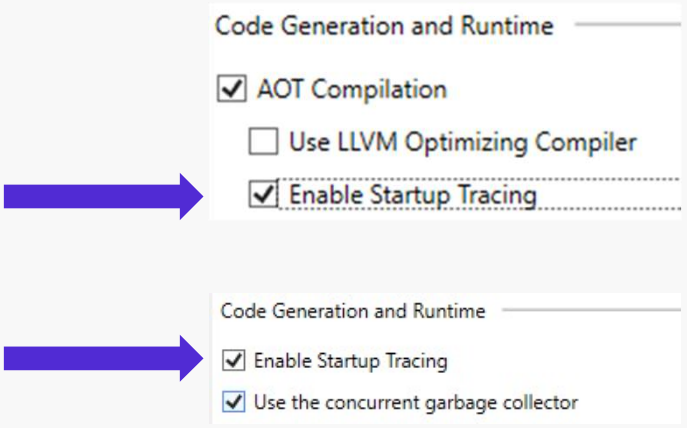
Innovations beyond native

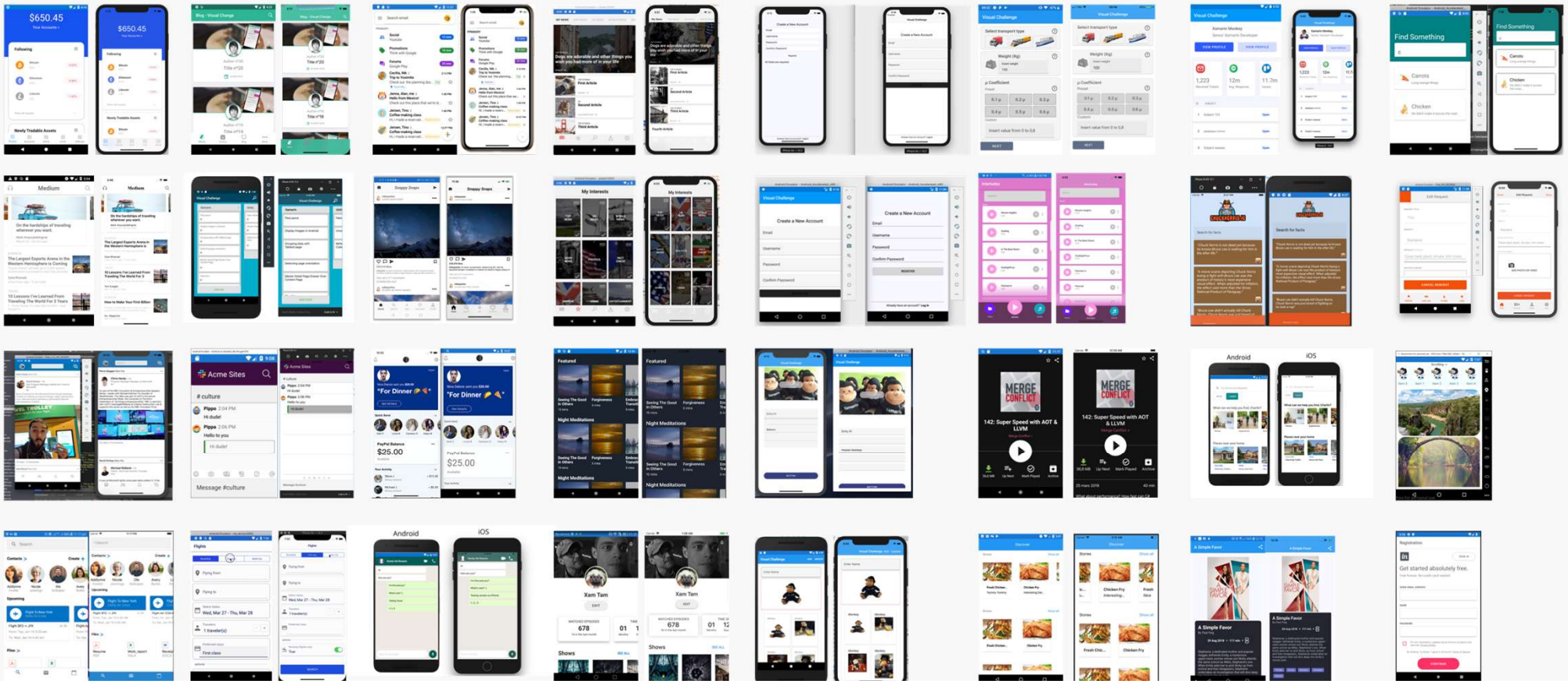
.NET



Android Startup Tracing

Type	Startup Speed	APK Size



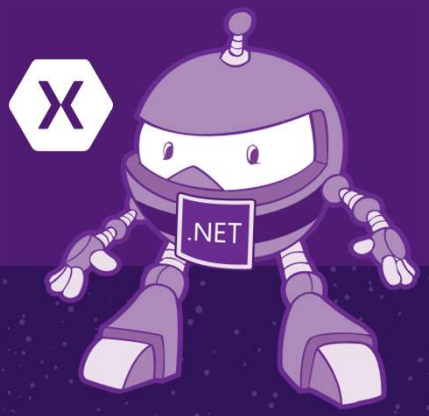

$650.45
Following
Blog - Visual Change
Title n°20
Title n°19
Search email
Social
Promotions
Forums
Google Play
Cecilia, Me 1
Trip to Yosemite
Check out the planning doc
Jenna, Alan, me 3
Hello from Mexico!
Jeroen, Tina 2
Coffee-making class
Hi, I made a mixed milk
Drops are adorable and other things you wish you had more of in your life
My Items
First Article
Second Article
Third Article
Fourth Article
Create a New Account
Email
Username
Password
Confirm Password
REGISTER
Already have an account? Log in
Visual Challenge
Select transport type
Weight (Kg)
μ Coefficient
Preset
Insert value from 0 to 0.8
NEXT
Xamarin Monkey
Senior Xamarin Developer
VIEW PROFILE
1,223
Reputation
12m
Avg. Response
11.7m
Hours
Find Something
Carrots
Long orange things
Chicken
You didn't make it across the road.
Medium
On the hardships of traveling wherever you want.
The Largest Esports Arena in the Western Hemisphere is Coming
10 Lessons I've Learned From Traveling The World For 3 Years
How to Make Your First Billion
Snappy Snaps
My Interests
Interfaces
Search for facts
Edit Request
CANCEL REQUEST
Acme Sites
culture
Pippa 2:04 PM
Hi dude!
Pippa 2:06 PM
Hello to you
Hi dude!
Message #culture
"For Dinner"
PayPal Balance
$25.00
Available
Featured
Swaying The Good in Others
Forgiveness
Night Meditations
MERGE CONFLICT
142: Super Speed with AOT & LLVM
Android
iOS
Flights
Flying from
Flying to
Select dates
Wed, Mar 27 - Thu, Mar 28
Travelers
1 traveler(s)
Preferred class
First class
SEARCH
Xam Tam
EDIT
WATCHED EPISODES
678
TIME
01
Shows
SEE ALL
Discover
Stories
Show all
Fresh Chicken...
Chicken Fry
Interesting...
A Simple Favor
Registration
Get started absolutely free.
First Name
Last Name
Email Address
Password
CONTINUE

Demo: Beautify Your Apps

.NET



Less time, less code



Xamarin.Essentials



Open Files

Secure Settings

Flashlight

Clipboard

Geolocation

Phone Dialer

Email

Device Info

Email Attachments

Accelerometer

Share

Geocoding

Vibration

File System

Connectivity

Share File

Sms

Preferences

Gyroscope

Magnetometer

Device Display Info

Battery

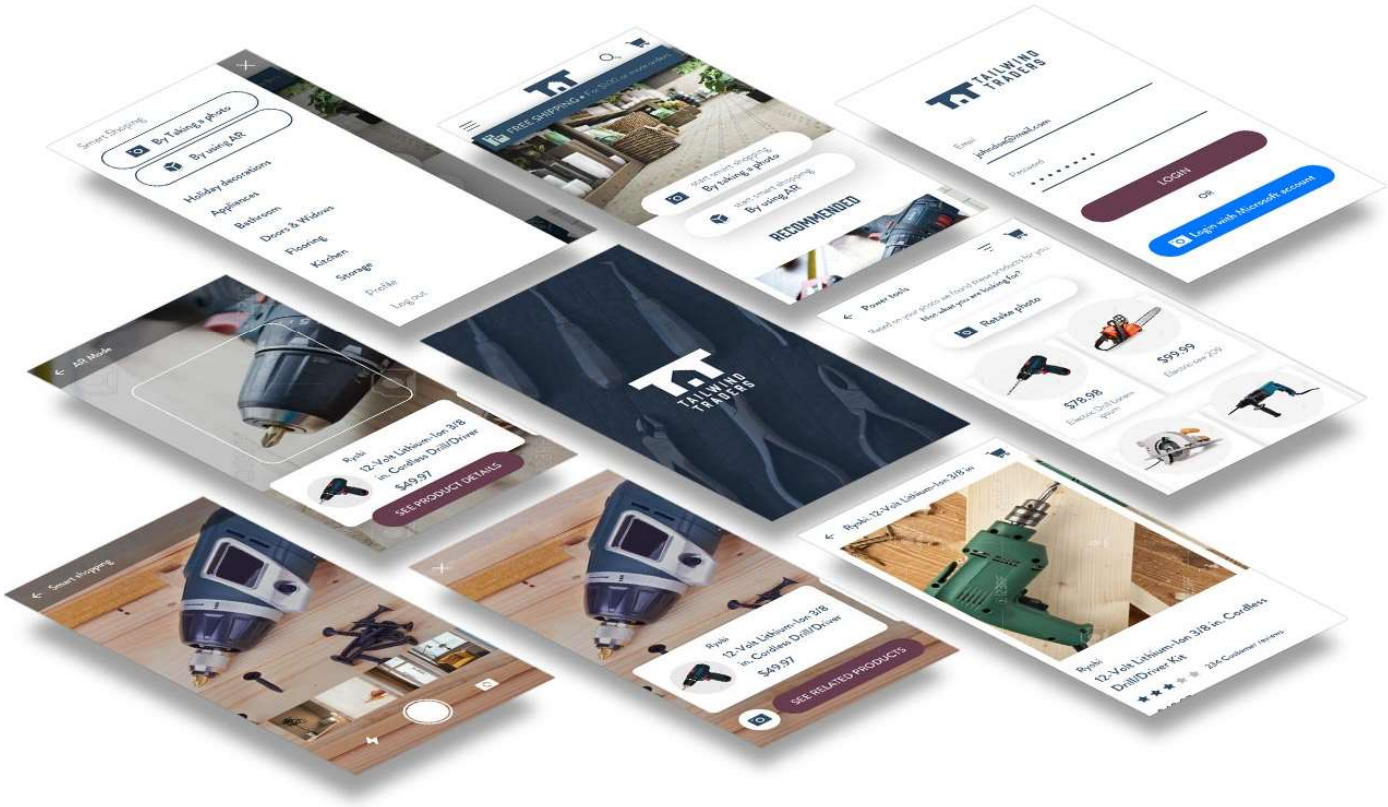
Compass

Screen Lock

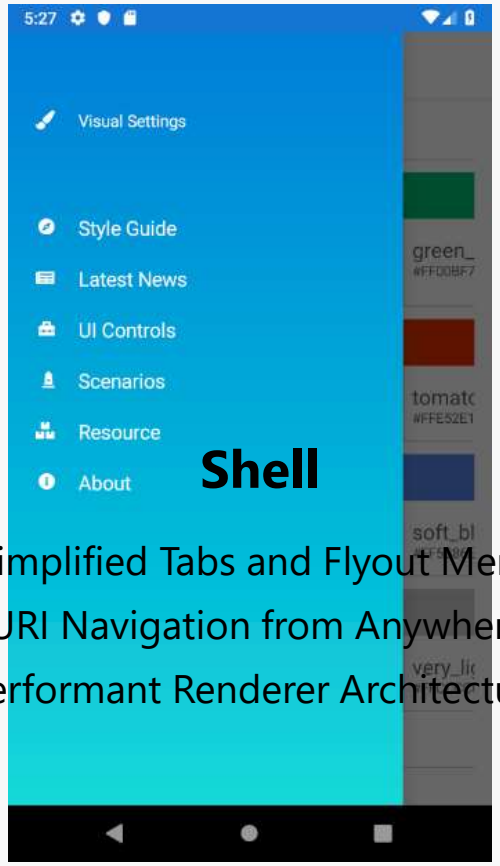
Text to Speech

Detect Shake

Xamarin.Forms



A Year in Xamarin.Forms

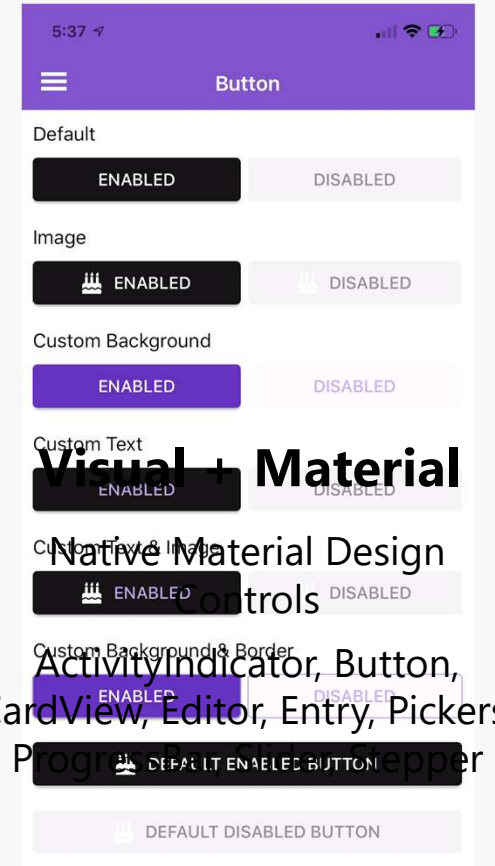


Shell

Simplified Tabs and Flyout Menu

URI Navigation from Anywhere

Performant Renderer Architecture



Visual + Material

Native Material Design Controls

ActivityIndicator, Button, CardView, Editor, Entry, Pickers, Progress Bar, Slider, Stepper



CollectionView

Linear, Grid, Custom Layouts

Performant Item Views

EmptyView

InfiniteLoading

Optimized Caching

A Year in Xamarin.Forms

Accessibility
BindableLayout EmptyView
C# Markup
Character Spacing
CheckBox
FontImageSource
GIF Animations
HTML Label
IndicatorView
Maps+
MediaElement
Modal Prompts
RadioButton
RefreshView
RelativeSource Binding
Shell Modals
SourceLink
VisualStateManager Target
WKWebView
+ Much More!

Flyout made simple

```
<MasterDetailPage xmlns="http://xamarin.com/schemas/2014/
            xmlns:x="http://schemas.microsoft.com/winfx/2
            xmlns:d="http:/
            xmlns:mc="http
            mc:Ignorable="
            xmlns:views="c
            x:Class="App6.
    <MasterDetailPage.Maste
        <views:MenuPage />
    </MasterDetailPage.Mast
    <MasterDetailPage.Detai
        <NavigationPage>
            <NavigationPage
                <OnPlatfor
                    <On Pl
                </OnPlatfor
            </NavigationPag
            <x:Arguments>
                <views:Iter
            </x:Arguments>
        </NavigationPage>
    </MasterDetailPage.Deta

</MasterDetailPage>
```

```
<FlyoutItem
    <ShellCo
</FlyoutItem
<FlyoutItem
    <ShellCo
</FlyoutItem
```

```
menuItems = new List<Hom
            {
                new Home
                new Home
            };

            ListViewMenu

            ListViewMenu
            ListViewMenu
            {
                if (e.Se
                    retu

                var id =
                await Ro
            };
```

```
public partial class MainPage : MasterDetailPage
    {
        Dictionary<int, NavigationPage> MenuPages = new Dictionary<int, NavigationPage>();
        public MainPage()
        {
            InitializeComponent();

            MasterBehavior = MasterBehavior.Popover;

            MenuPages.Add((int)MenuItemType.Browse, (NavigationPage)Detail);
        }

        public async Task NavigateFromMenu(int id)
        {
            if (!MenuPages.ContainsKey(id))
            {
                switch (id)
                {
                    case (int)MenuItemType.Browse:
                        MenuPages.Add(id, new NavigationPage(new ItemsPage()));
                        break;
                    case (int)MenuItemType.About:
                        MenuPages.Add(id, new NavigationPage(new AboutPage()));
                        break;
                }
            }

            var newPage = MenuPages[id];

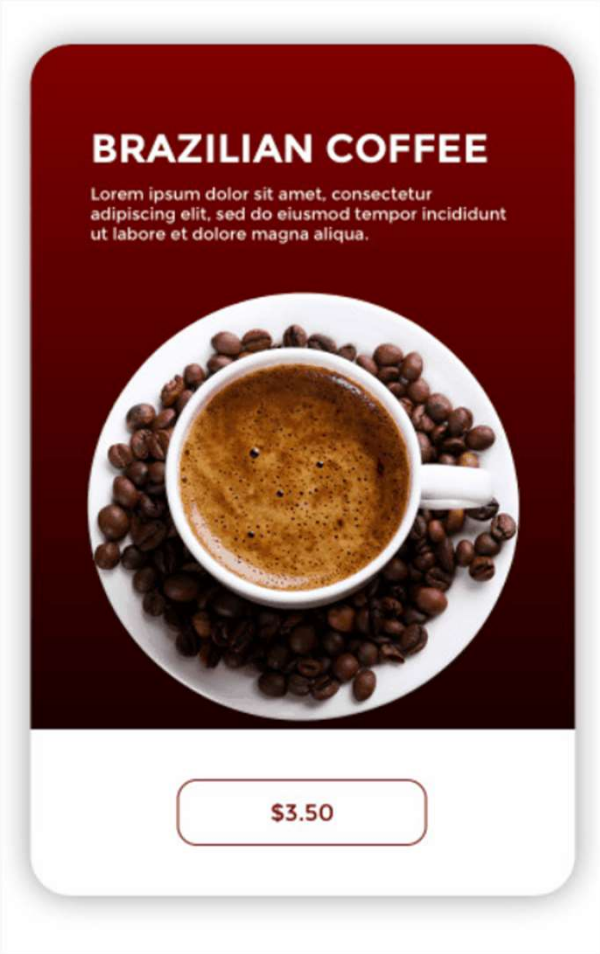
            if (newPage != null && Detail != newPage)
            {
                Detail = newPage;

                if (Device.RuntimePlatform == Device.Android)
                    await Task.Delay(100);

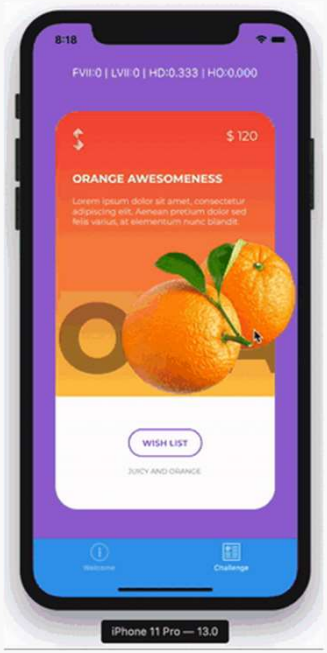
                IsPresented = false;
            }
        }
    }
```

```
        </StackLayout>

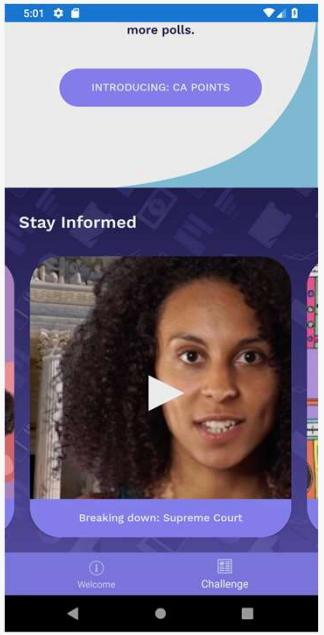
    </ContentPage>
```

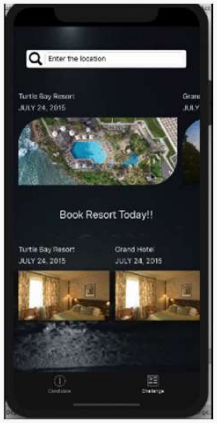


CarouselView

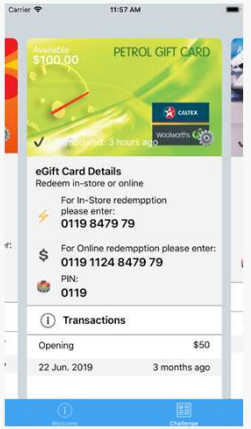


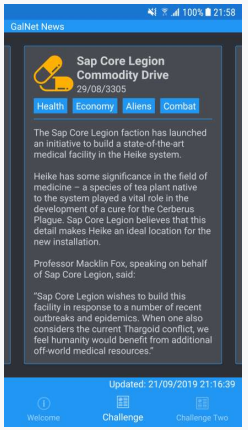






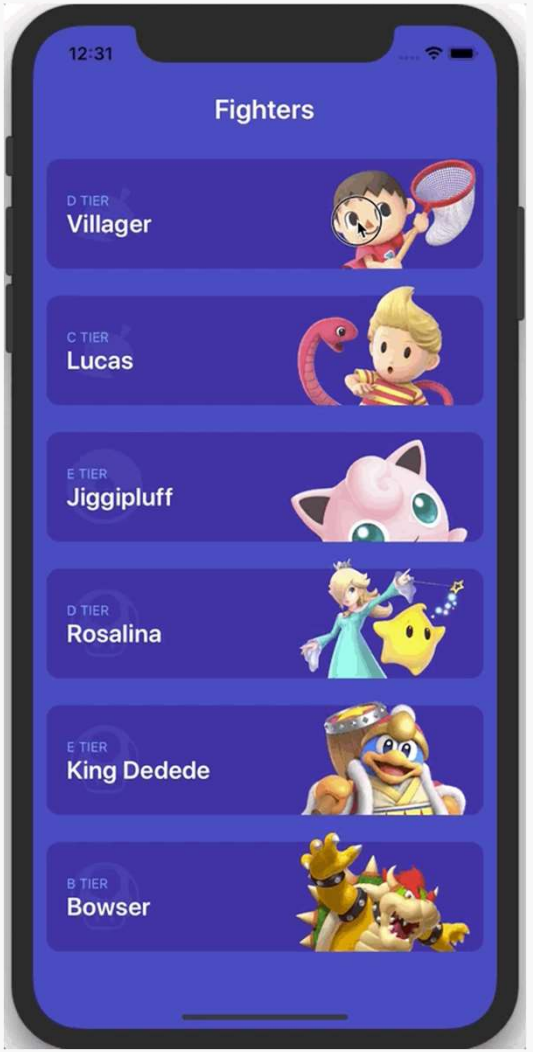






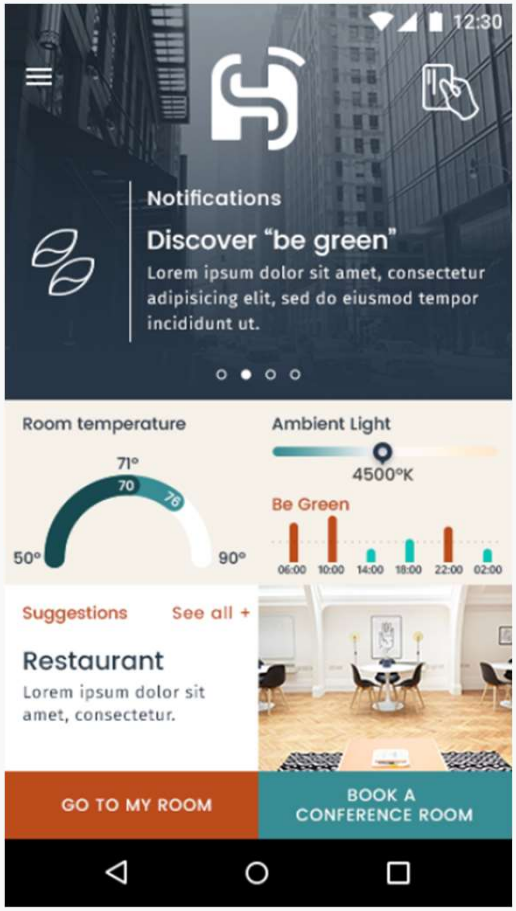
Xamarin.Forms 4.5.0

- SwipeView
- VisualStateManager Target
- StateTriggers
- Media Element
- Custom Embedded Fonts
- Platform Specifics
  - Image location
  - Picker UpdateMode
  - Android Back Button Color



Faster builds for faster development

Step	VS 2017 (15.8)	VS 2019 (16.0)	Delta
First Build	01:04.20	00:50.13	-21.95%
Incremental Build (XAML Change)	00:10.62	00:07.47	-29.66%
Deploy (XAML Change)	00:09.03	00:04.44	-50.83%



XAML Hot Reload for Xamarin.Forms



aka.ms/XAMLHotReloadDocs

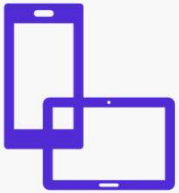
Debug your app, make a change, and hit save. See your changes instantly.



No setup or code changes



Interact with your real data



Android or iOS



Emulator, Simulator, or Device



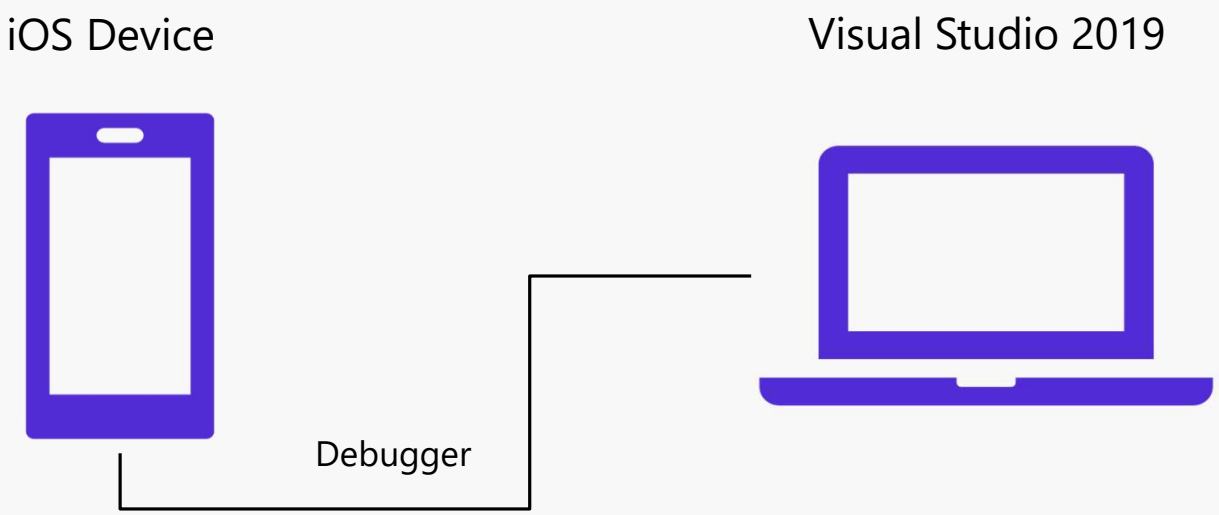
Intelligent and resilient



Works with custom controls, third party libraries and frameworks, Shell, resources, and more!

Xamarin Hot Restart

Restart your app with your changes, and without waiting for packaging and deployment.

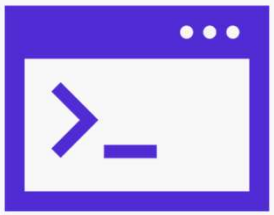




aka.ms/HotRestartDocs

Android Apply Changes

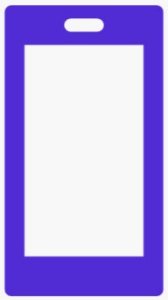
Restart your app using native Android development technology in Visual Studio.



XML Changes

Package diff





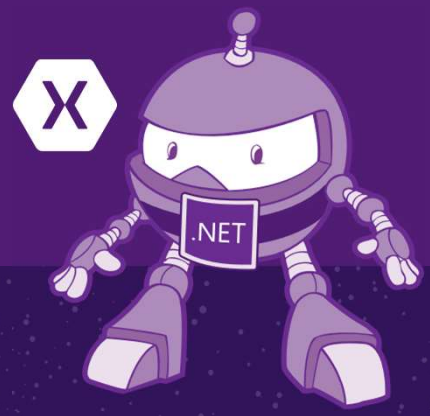
Running Android app
(Activity restarts)

Build less, do more

.NET



Celebrating you!





aka.ms/xamstreamers



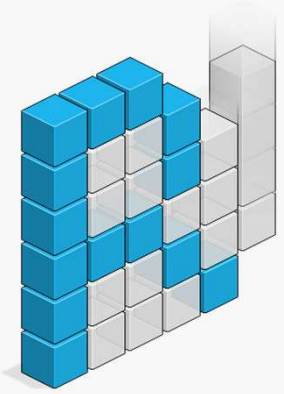

Frameworks



MVVMCross



Uno



ReactiveUI



Reactive Extensions






Libraries & Tools



Cognitive Services





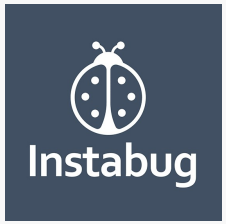




Azure Active Directory B2C









Zxing Barcode Scanning

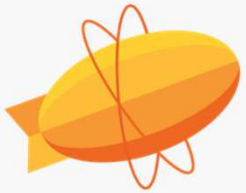


Akavache



Gorilla Player





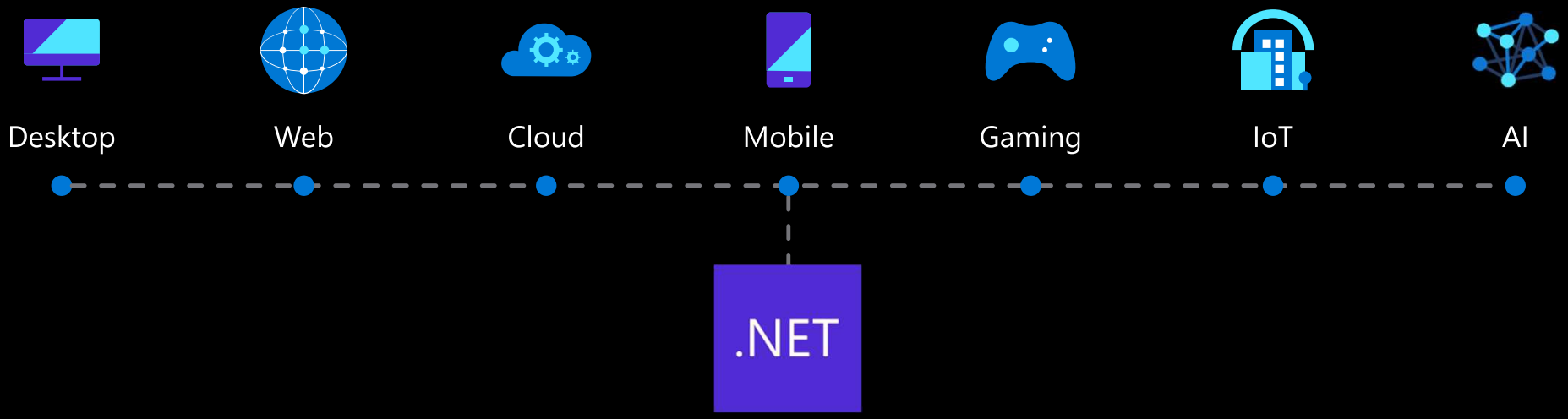
Zeplin




Thank you!
Zombie

.NET

Your platform for building anything



Ask your questions LIVE on Twitter

#dotNETConf



If we select your question, we'll send you a free Xamarin plush monkey!

